

Algorithms and Programs

1. Conditional Operator (Voting Eligibility)

Algorithm:

- Step 1: Start the program
- Step 2: Declare an integer variable age
- Step 3: Read the value of age from the user
- Step 4: Check the condition (age >= 18) using the conditional (ternary) operator
- Step 5: If the condition is true, print "You are eligible to vote."
- Step 6: If the condition is false, print "You are not eligible to vote."
- Step 7: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int age;

    printf("Enter your age: ");
    scanf("%d", &age);

    (age >= 18) ? printf("You are eligible to vote.\n")
                : printf("You are not eligible to vote.\n");

    getch();
}
```

2. if (Password Check)

Algorithm:

- Step 1: Start the program
- Step 2: Declare an integer variable password
- Step 3: Read the value of password from the user
- Step 4: Check the condition (password == 1234)
- Step 5: If the condition is true, print "Access granted."
- Step 6: If the condition is false, skip the print statement (no output)
- Step 7: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int password;

    printf("Enter your password: ");
    scanf("%d", &password);

    if (password == 1234) {
        printf("Access granted.\n");
    }

    getch();
}
```

3. switch (Recharge Plan Selector)

Algorithm:

- Step 1: Start the program
- Step 2: Declare an integer variable plan
- Step 3: Display the list of available plans (Rs. 199, Rs. 299, Rs. 499, Rs. 599, Rs. 999)
- Step 4: Read the user's choice into plan
- Step 5: Check the value of plan using a switch statement
- Step 6: If plan is 1, print "You selected Rs. 199 Plan." and exit using break
- Step 7: If plan is 2, print "You selected Rs. 299 Plan." and exit using break
- Step 8: If plan is 3, print "You selected Rs. 499 Plan." and exit using break
- Step 9: If plan is 4, print "You selected Rs. 599 Plan." and exit using break
- Step 10: If plan is 5, print "You selected Rs. 999 Plan." and exit using break
- Step 11: If plan does not match any case, print "Invalid plan selected." (default)
- Step 12: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int plan;

    printf("1. Rs. 199 Plan\n");
    printf("2. Rs. 299 Plan\n");
    printf("3. Rs. 499 Plan\n");
    printf("4. Rs. 599 Plan\n");
    printf("5. Rs. 999 Plan\n");
    printf("Enter your choice: ");
    scanf("%d", &plan);

    switch (plan) {
        case 1:
            printf("You selected Rs. 199 Plan.\n");
            break;
        case 2:
            printf("You selected Rs. 299 Plan.\n");
            break;
        case 3:
            printf("You selected Rs. 499 Plan.\n");
```

```
        break;
    case 4:
        printf("You selected Rs. 599 Plan.\n");
        break;
    case 5:
        printf("You selected Rs. 999 Plan.\n");
        break;
    default:
        printf("Invalid plan selected.\n");
}

getch();
}
```

4. goto (Counting Program)

Algorithm:

Step 1: Start the program

Step 2: Declare an integer variable i and initialize it to 1

Step 3: Define a label named start

Step 4: Print the value of i

Step 5: Increment the value of i by 1

Step 6: Check the condition (i <= 5)

Step 7: If the condition is true, jump back to the label start using goto and repeat from Step 4

Step 8: If the condition is false, continue to the next step

Step 9: Stop the program

Program:

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main() {
```

```
    int i = 1;
```

```
    start:
```

```
        printf("Count: %d\n", i);
```

```
        i++;
```

```
        if (i <= 5) {
```

```
            goto start;
```

```
        }
```

```
        getch();
```

```
}
```

5. for Loop

Algorithm:

- Step 1: Start the program
- Step 2: Declare an integer variable i
- Step 3: Initialize i to 1 in the for loop
- Step 4: Check the condition ($i \leq 5$)
- Step 5: If true, print the value of i
- Step 6: Increment i by 1
- Step 7: Repeat Steps 4 to 6 until the condition becomes false
- Step 8: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int i;

    for (i = 1; i <= 5; i++) {
        printf("%d\n", i);
    }

    getch();
}
```

6. while Loop

Algorithm:

Step 1: Start the program

Step 2: Declare an integer variable i and initialize it to 1

Step 3: Check the condition ($i \leq 5$)

Step 4: If true, print the value of i

Step 5: Increment i by 1

Step 6: Repeat Steps 3 to 5 until the condition becomes false

Step 7: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int i = 1;

    while (i <= 5) {
        printf("%d\n", i);
        i++;
    }

    getch();
}
```

7. do-while Loop

Algorithm:

- Step 1: Start the program
- Step 2: Declare an integer variable i and initialize it to 1
- Step 3: Print the value of i
- Step 4: Increment i by 1
- Step 5: Check the condition ($i \leq 5$)
- Step 6: If true, repeat from Step 3
- Step 7: If false, stop the loop
- Step 8: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int i = 1;

    do {
        printf("%d\n", i);
        i++;
    } while (i <= 5);

    getch();
}
```

8. Matrix Addition and Subtraction

Algorithm:

Step 1: Start the program

Step 2: Declare two 2x2 matrices a and b, and two 2x2 matrices sum and diff to store the results

Step 3: Declare integer variables i and j for loop control

Step 4: Read the elements of the first matrix a using nested loops

Step 5: Read the elements of the second matrix b using nested loops

Step 6: For each position [i][j], calculate $\text{sum}[i][j] = a[i][j] + b[i][j]$

Step 7: For each position [i][j], calculate $\text{diff}[i][j] = a[i][j] - b[i][j]$

Step 8: Print all elements of the sum matrix, row by row

Step 9: Print all elements of the diff matrix, row by row

Step 10: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int a[2][2], b[2][2], sum[2][2], diff[2][2];
    int i, j;

    printf("Enter elements of first 2x2 matrix:\n");
    for (i = 0; i < 2; i++)
        for (j = 0; j < 2; j++)
            scanf("%d", &a[i][j]);

    printf("Enter elements of second 2x2 matrix:\n");
    for (i = 0; i < 2; i++)
        for (j = 0; j < 2; j++)
            scanf("%d", &b[i][j]);

    for (i = 0; i < 2; i++) {
        for (j = 0; j < 2; j++) {
            sum[i][j] = a[i][j] + b[i][j];
            diff[i][j] = a[i][j] - b[i][j];
        }
    }

    printf("Sum of matrices:\n");
```

```
    for (i = 0; i < 2; i++) {  
        for (j = 0; j < 2; j++)  
            printf("%d ", sum[i][j]);  
        printf("\n");  
    }  
  
    printf("Difference of matrices:\n");  
    for (i = 0; i < 2; i++) {  
        for (j = 0; j < 2; j++)  
            printf("%d ", diff[i][j]);  
        printf("\n");  
    }  
  
    getch();  
}
```

9. String Operations Using Library Functions

Algorithm:

- Step 1: Start the program
- Step 2: Declare two character arrays str1 and str2
- Step 3: Read a string into str1 from the user
- Step 4: Find the length of str1 using strlen() and print it
- Step 5: Copy the contents of str1 into str2 using strcpy()
- Step 6: Print the copied string str2
- Step 7: Concatenate "_end" onto str1 using strcat()
- Step 8: Print the concatenated string str1
- Step 9: Compare str1 and str2 using strcmp()
- Step 10: If the comparison result is 0, print "Strings are equal"
- Step 11: If the comparison result is not 0, print "Strings are not equal"
- Step 12: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>
#include <string.h>

void main() {
    char str1[50], str2[50];

    printf("Enter a string: ");
    scanf("%s", str1);

    printf("Length of string: %d\n", strlen(str1));

    strcpy(str2, str1);
    printf("Copied string: %s\n", str2);

    strcat(str1, "_end");
    printf("After concatenation: %s\n", str1);

    if (strcmp(str1, str2) == 0)
        printf("Strings are equal\n");
    else
        printf("Strings are not equal\n");
    getch();
}
```

}

10. Recursion (Factorial)

Algorithm:

- Step 1: Start the program
- Step 2: Declare an integer variable num and a function factorial(int n)
- Step 3: Read the value of num from the user
- Step 4: Call factorial(num)
- Step 5: Inside the function, check if $n == 0$
- Step 6: If true, return 1 (base case)
- Step 7: If false, return $n * \text{factorial}(n - 1)$ (recursive case)
- Step 8: Print the returned result
- Step 9: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

int factorial(int n) {
    if (n == 0) {
        return 1;
    }
    return n * factorial(n - 1);
}

void main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Factorial of %d is %d\n", num, factorial(num));

    getch();
}
```

11. Print a Number in Reverse Order

Algorithm:

- Step 1: Start the program
- Step 2: Declare integer variables num, rev (initialized to 0), and digit
- Step 3: Read the value of num from the user
- Step 4: Check the condition (num != 0)
- Step 5: If true, extract the last digit using digit = num % 10
- Step 6: Update rev = rev * 10 + digit
- Step 7: Update num = num / 10
- Step 8: Repeat Steps 4 to 7 until num becomes 0
- Step 9: Print the value of rev
- Step 10: Stop the program

Program:

```
#include <stdio.h>
#include <conio.h>

void main() {
    int num, rev = 0, digit;

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num != 0) {
        digit = num % 10;
        rev = rev * 10 + digit;
        num = num / 10;
    }

    printf("Reversed number: %d\n", rev);

    getch();
}
```